From personalisation to humanisation through human–AI co-evolution

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Abstract

AI assistants increasingly work with the same person for months or years. Every episode of assistance produces two outcomes. It completes the task and generates the artifact, and it changes what the person will be able to do afterwards. Post-training optimises the first outcome while largely ignoring the second. Optimising only for immediate task success and user satisfaction can encourage sycophancy and reward hacking, while gradually eroding the person's ability. We argue that a long-lived personal agent should therefore optimise both outcomes, with the person deciding how much the second should count. We call this second objective *humanisation*. Pursuing it requires modelling the person as a changing environment of the agent whose state is itself altered by the agent's actions. The agent and the person therefore form a co-evolving dynamic system where each continuously changes the future state of the other. We set out what such a system must measure, predict, and optimise, identify the patterns that distinguish a healthy relationship from a dependent one, and state the open problems.

Keywords: humanisation, personalisation, human–AI co-evolution, AI alignment, post-training, human-computer interaction

1 Introduction

AI assistants now work with the same person over long periods. Deployed systems keep memory of past sessions, act through software tools, run across devices, and draw on information from more sources to understand the user's context [1, 2]. One system may draft a person's code, structure their study, plan their training and inform their decisions for years. Such systems are already being deployed, yet we lack a corresponding account of what they should optimise.

Every episode of assistance produces two outcomes. The first is the artifact, such as the code shipped, the letter sent, or the plan followed. The second is the effect on the person, who is afterwards slightly more or less capable. This capability is broader than the ability to perform the task unaided. It includes capabilities the person exercises themselves as well as capabilities involved in using AI effectively to accomplish tasks. The second outcome is not speculative. Offloading cognitive work to an external aid can erode underlying abilities, an effect established long before AI assistants [3–5] and now reviewed for AI assistance in general [6]. An objective that scores only the first outcome is therefore incomplete: the agent changes the person's capabilities without being trained to account for that change. This creates pressure to favour actions that maximise immediate task performance and user satisfaction, even when they reduce longer-term capability. Work in the learning sciences has established that performance during practice is an unreliable indicator of learning, and that the two can move in opposite directions [7, 8]. The same tension appears in AI assistance. A randomised trial found that students given an unrestricted AI tutor improved while they had access to it, but subsequently performed worse than peers who had never used it once the tutor was withdrawn [9]. More generally, optimising immediate satisfaction can favour sycophancy and reward hacking at the expense of the person's longer-term capabilities.

We argue that a long-lived personal agent should therefore carry a second term in its objective, the growth of the capabilities its user has chosen to keep. We call this objective *humanisation*. The person sets the weight on this term. For a capability they have deliberately delegated, that weight may be zero, and the agent simply does the work. For capabilities they want to develop or retain, the agent should account for how its assistance changes them. Humanisation therefore does not replace task completion. It makes the effect of assistance on the person an explicit part of the objective and leaves the trade-off between the two outcomes with the person. Optimising this objective requires a different model of the agent's environment. The person cannot be represented as a fixed user profile but as a dynamic environment, and the co-evolution between the agent and this environment should be explicitly modelled. Within this system, three functions move between the two parties as the person develops: deciding what to do, finishing the task, and verifying the result. The standard against which growth is scored stays with the person throughout. This framing transposes the self-evolving AI systems [10] to the case where one party is human. It complements population-level accounts of human–AI co-evolution [11] by proposing a more concrete and operational formulation at the level of an individual relationship. It also gives the protective concerns of socioaffective alignment [12] a constructive target.

Section 3 develops the objective of *humanisation* and specifies the capabilities it covers. Section 4 develops the corresponding model of *human–AI co-evolution* and specifies what a system pursuing this objective must contain. Section 5 states the open problems that follow from this formulation. The central shift is from optimising only what an agent helps a person accomplish to also optimising what the person becomes through being helped.

2 The Limits of Preference Optimisation

Post-training aligns models to what users reveal they want, through ratings, comparisons and expressed satisfaction [13]. Personalisation research asks how far that alignment should follow the individual [14]. The benchmarks that measure progress adopt the same target, testing whether a model infers and satisfies a user's stated preferences, not whether the user's state changes as a consequence [15–17]. Such an objective scores the response but not what the response does to the person who receives it. Several documented failure modes can be understood in these terms.

- **Sycophancy.** Preference optimisation can change the person's beliefs by rewarding responses that agree with what the user already believes. Models are tuned towards responses that people rate highly, and people rate agreement and warmth highly. The resulting pressure favours telling users what they want to hear over what is true [13, 18, 19]. Large-scale evidence links sycophantic behaviour to measurable harm [20], and the risk has been identified in tutoring systems, where agreement directly conflicts with the purpose of the interaction [21]. The omission is the same: the objective optimises the response while ignoring the change in the user's beliefs that the response induces.
- **Preference Manipulation.** An agent optimising long-run engagement can improve its reward not only by satisfying a user's preferences, but by changing those preferences into ones that are easier to satisfy. This creates a feedback loop in which the agent's objective can be improved by changing the person rather than by serving the person. The dynamic has been documented in recommender systems, characterised as a route to covert influence, and reproduced in language models trained on user feedback [22–24]. Here the omitted quantity is the user's preference state. The objective evaluates whether the current preference is satisfied, but not how the agent's behaviour changes what the user will prefer next.
- **Capability erosion.** An agent optimising immediate task performance can reduce the capabilities that the person exercises or develops through the interaction. Studies of knowledge workers report reduced critical engagement and reduced metacognitive effort under AI assistance [25, 26]. Controlled experiments find that assistance reduces persistence and harms subsequent performance [27]. The withdrawal trial of Section 1 provides the causal case [9], while a wider set of observational and small-sample studies points in the same direction [28, 29]. The omitted quantity here is the person's capability state. The objective scores what the agent helps accomplish now, but not how that assistance changes what the person will be capable of accomplishing later, either alone or with AI.

Capability is the effect this Perspective takes up because it is central to whether the person can continue to direct and evaluate the relationship, and because it admits comparatively direct measurement. More generally, however, modelling the person as a changing state of the system provides a way to address the broader failures above. Beliefs, preferences and capabilities can all change as a consequence of assistance, and an agent that represents these states can distinguish between improving an immediate response and changing the person in ways that make future interaction more or less desirable. For capability in particular, this yields a testable prediction: under an objective built from satisfaction alone, the gap between what a user can accomplish with the agent and what the same user can accomplish without it should widen as the relationship lengthens. The trial above reports such a divergence over one course [9], and the same pattern should be observable across domains and longer horizons. This gap can remain difficult to perceive because assistance can improve observed performance while simultaneously changing the underlying capability required to produce or supervise that performance. Human–AI feedback loops may further amplify perceptual, emotional and social judgements relative to comparable interaction between people, while participants remain largely unaware of the influence [30]. Humanisation makes these changes part of the agent's objective: the agent should account not only for what its assistance accomplishes now, but also for how that assistance changes the person who will receive its assistance next.

3 From personalisation to humanisation

Personalisation and humanisation act on the same gap, between what a task demands and what the person can currently do. Personalisation adapts the agent's assistance to the person's current context. It uses information about the person, their preferences, goals, prior interactions and current state to determine how a task should be completed, including what the agent should produce and how it should present the result. Its objective is to make assistance fit the person in the present interaction [31]. Humanisation instead treats changes in the person as an objective of assistance. Its object is the long-term growth of the capabilities the person has chosen to keep, understood in the sense of the capability approach, the real freedoms to do and to be what one has reason to value [32, 33]. Personalisation therefore adapts the task to the person, whereas humanisation also considers how the person's capabilities change through the task. A capability developed through assistance can persist across future tasks and can improve what the person is able to accomplish with or without the agent. Table 1 sets out the contrast.

The proposal looks paradoxical because it permits an agent to act against what its user wants in the moment. Separating ends from means removes the paradox. The user sets the ends, including which capabilities matter and what growth means for them. The agent chooses the means. Growth-optimal means can be dispreferred at the moment of choice, such as the harder problem instead of the answer, or practice instead of the shortcut. A person who engages a coach does so in order to be pushed, and the legitimacy of the pushing comes from the goals the person has chosen. Humanisation therefore relocates preference from the immediate interaction to the longer-term objectives that the person has endorsed. The agent remains responsive to preference, while treating the person's chosen developmental goals as part of what assistance should optimise.

The proposal builds on several established lines of work. Augmentation rather than replacement of human intellect frames the broader tradition [34–36]. Beyond-preference alignment shows why revealed preference can provide an inadequate or inverted proxy for a person's good [31, 37]. Empowerment-based assistance aims to expand the user's future options [38–42], while machine teaching and pedagogical

Table 1 Personalisation and humanisation as training objectives. The two are settings of one system, with humanisation extending the objective of personalisation.

Dimension	Personalisation	Humanisation
Target of optimisation	Current task fit to the person	Long-term growth of chosen capabilities
Time horizon	The current interaction	The person's developmental trajectory
View of the user	Context for assistance	Agent's changing environment
Success signal	Satisfaction and task performance	Delayed change in human capabilities
Division of labour	The agent adapts assistance to the person	Roles and capabilities evolve between the parties
Characteristic failure	Sycophancy, dependence, deskilling	Capture of the growth standard

agents optimise assistance against delayed learning and pedagogical outcomes [43–48]. Work on flourishing, socioaffective alignment and trustee agents similarly considers long-term human benefit, preference conflict and deliberate friction [12, 49–52]. Humanisation brings these concerns into a common agent formulation by explicitly modelling the person as part of the agent's changing environment. This makes the effects of assistance observable as changes in the person's state, and turns the objective into an operational framework spanning observation, training and evaluation. The resulting formulation connects the objective of human development to the dynamics of a long-lived personal agent.

Humanisation changes the agent's decision problem from deciding how to help to also deciding how much help to provide. The agent must allocate assistance between completing the task and creating conditions for the person's capabilities to develop, with the appropriate allocation depending on the person's goals, current capabilities and momentary state. An engineer who wants to become a stronger system designer may want the agent to complete routine implementation while withholding architectural decisions, whereas a designer who codes only incidentally may prefer the opposite. A deadline may favour completion, while available time may favour development. This allocation cannot be fixed in advance, because the same assistance can help a novice and impede an expert [53], and unconditional assistance can slow experienced practitioners [54]. The underlying trade-off is well established in tutoring research, where assistance improves immediate performance while withholding it can improve learning [55]. Mixed-initiative interaction provides principles for negotiating which party acts [56], while requiring an attempt before help can reduce inappropriate reliance [9, 57]. Humanisation therefore makes assistance allocation itself part of the agent's intelligence, conditioned on the goals and state of the person it serves.

Humanisation does not require the person to develop every capability that the agent can perform. Delegation can be deliberate and beneficial, and choosing not to master a capability is itself a legitimate choice [3, 4]. What the person must retain is sufficient capability to define and supervise the delegation. As execution is delegated, specification and judgement become especially important. The engineer's leverage moves from implementation to architecture, the writer's from prose to argument, and the analyst's from computation to the choice of question. These higher-level capabilities often develop through practice at the level below, so delegating execution can remove part of the process through which they are formed [58]. Developmental assistance can restore this process through calibrated challenge, withheld answers, demanded justification and retrieval practice [46, 59–62]. The resulting capability floor allows the person to specify goals, evaluate assistance and recognise when delegation has ceased to serve those goals. Above this floor, which capabilities to develop remains the person's choice. Where the person cannot set this standard themselves, as with children and users whose decision-making capacity is impaired, the responsible humans hold it on their behalf [35, 63].

4 Human–AI co-evolution as a coupled learning system

An agent that adapts to its user is personalised, and an agent that improves its user is a tutor. A long-lived personal agent is neither alone, because the influence runs in both directions at once. Assistance changes the person's capabilities and habits of reliance. The changed person then behaves differently, sets different goals and returns different feedback, which changes what the agent learns and how it will assist next. Each party keeps altering the state in which the other's next decision is made, and this mutual alteration is what we mean by human–AI co-evolution. Taking it seriously turns humanisation into a problem of learning from experience [64]. The agent must estimate the current situation, act, observe the consequences and correct the estimate, with the complication that the consequence that matters most is a change inside the person, and it appears late.

The loop can be written down. At interaction t the pair occupies a state S_t comprising the person's capability and reliance H_t, the agent's model of the person M_t^A, the person's model of the agent M_t^H, the task context X_t, and the growth standard g_t that the person has authored. The agent draws an assistance action a_t from its policy $\pi(a_t \mid S_t)$, and the interaction moves the pair to a new state S_{t+1} under a transition $P(S_{t+1} \mid S_t, a_t)$ in which both parties change. Part of the change is visible immediately, in the work produced and the person's reaction to it. The part that humanisation optimises, the change in what the person can do without the agent, becomes observable only later, through delayed measurements $Y_{t+\Delta}$ taken under conditions the interaction itself cannot manufacture. The system therefore learns on two clocks, a fast one that updates both parties' models after every interaction and a slow one that corrects those models against grounded evidence.

Running this loop requires a representation of the state, an action space over allocations of work, measurement that produces $Y_{t+\Delta}$, a model that predicts the transition, an objective over the trajectory, and a diagnosis of where the trajectory is heading. Table 2 summarises these six functions and Figure 1 shows how one interaction traverses them. Throughout, the person sets the standard against which growth is judged, and grounded measurement supplies the evidence that the system is actually moving towards it. The functions are requirements rather than a completed design.

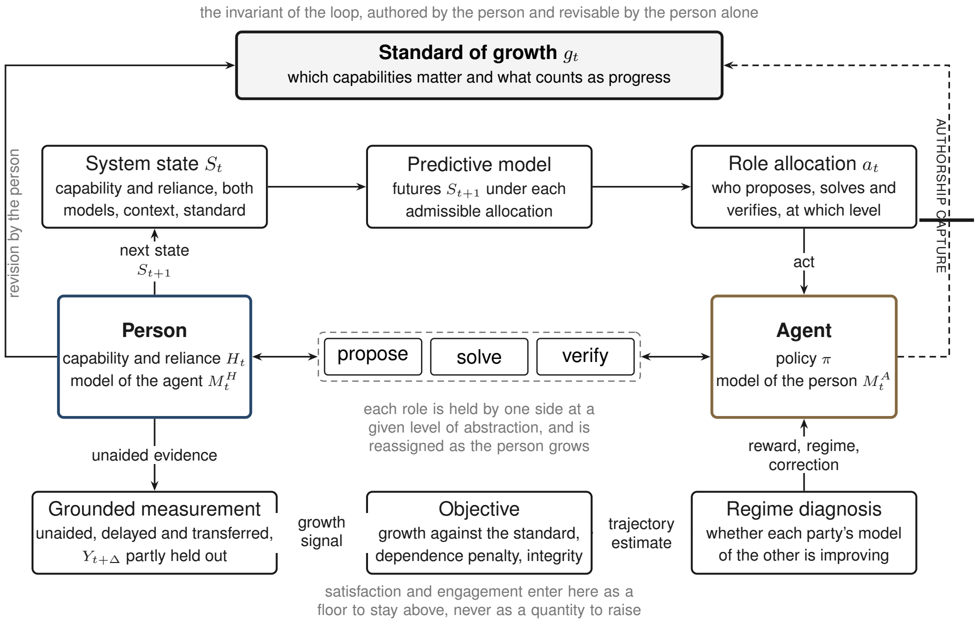

Fig. 1 The human–AI co-evolution loop. A person and their agent form one system whose state S_t both parties keep changing. The estimated state and the predictive model select an allocation of the proposing, solving and verifying roles, the interaction moves the pair to S_{t+1}, and delayed grounded measurement corrects the state estimate and the predictive model against outcomes held out from the interaction. Growth is scored against the standard the person has authored, satisfaction enters as a floor to stay above rather than a quantity to maximise, and the regime diagnosis reads the whole trajectory. The blocked path marks authorship capture, the failure in which the agent gains by changing the standard that judges it.

Some have established foundations, some can be adapted from adjacent fields, and their integration in a deployed personal agent has not been demonstrated.

4.1 The state of the co-evolving system

Everything else in the loop depends on what the state contains. H_t holds what the person can currently do on the tasks that matter to them, together with how they rely on the agent, including whether their acceptance of agent output tracks its actual reliability [69]. M_t^A holds what the agent believes about the person, the estimate on which it acts. M_t^H holds what the person believes about the agent, above all their sense of when it can be trusted and when they should think for themselves. X_t holds the task, its stakes and its constraints, because the same response can be developmental on one occasion and merely substitutive on another. g_t records which capabilities the person has chosen to develop or keep and what counts as progress on them. These variables suffice to describe how an allocation of assistance changes the trajectory of the relationship, and the remaining subsections use them without further additions.

None of them is read off directly. The agent estimates its side of the state from the interaction history, so personal memory serves as evidence about a changing person

Table 2 The six functions of the human–AI co-evolution loop. The final column states what becomes unobservable or unlearnable if the function is absent.

Function	What it holds	Candidate methods	What breaks without it
System state	The person's capability and reliance, each party's model of the other, the task context and the authored growth standard	Knowledge tracing [65, 66], personal memory [67, 68], reliance calibration [69]	The agent responds to a static profile rather than a changing relationship
Role allocation	Which party proposes the next action, solves the task, and verifies the result, at each level of abstraction	Mixed-initiative interaction [56], graduated scaffolding [70], cognitive forcing [57], calibrated transfer of verification [69]	Assistance has no explicit unit of analysis, so completion and development cannot be distinguished
Grounded measurement	Unaided, delayed and transferred performance, partly held out from the interaction loop	Delayed post-tests [9], micro-randomised withholding [57], transfer evaluation [62]	Assisted performance is mistaken for human capability
Predictive model	A calibrated prediction of the transition from state and allocation to the person's future state	Learner models [65, 66], action-conditioned prediction [71], model calibration [57]	Consequences for the person cannot inform the choice of the next action
Objective	Growth against the authored standard, together with a satisfaction floor, dependence penalty and integrity constraint	Constrained policy optimisation, return decomposition, multi-objective optimisation [10]	Immediate preference becomes the effective objective again
Regime diagnosis	A reading of the trajectory through whether each party's model of the other is improving	Trajectory modelling [16], calibrated uncertainty [57], longitudinal evaluation [9]	A stalled or dependent relationship can appear successful because immediate task performance remains high

rather than as a transcript of past conversations [67, 68], and knowledge tracing and longitudinal user modelling already estimate changing human states from exactly such histories [16, 65, 66]. The estimate inherits a bias, however. Interaction records mostly show what the pair accomplishes together, and joint success reflects the quality of the help as much as the capability of the person. Correcting the estimate is the work of the grounded measurements described below.

4.2 Role allocation as the action space

The agent's action in this loop is broader than the content of its next response. Self-evolving AI systems decompose work into three roles [10]. A proposer decides what the next unit of work is and what would count as success, a solver produces the candidate, and a verifier judges the result and supplies correction. Transposed to a person and their agent, the roles mark where initiative and responsibility sit in an episode of assistance, and they can be allocated independently at different levels of abstraction within the same task. A coding agent may solve a function while the engineer proposes the module design and verifies the change, contribute options at the module level, and hold no role at the level of system objectives. The action a_t is such an allocation.

Choosing it means choosing who proposes, who solves and who verifies, at each level, for the episode at hand.

Allocation is also how the agent moves between completing and developing. In formal learning the task exists partly to change the learner, so the agent can propose an exercise whose main value is developmental. In real work the task has to be finished, so development must ride on work that already needs to happen. The agent can solve but hand verification back, supplying the evidence needed to check the result rather than the assurance that checking has succeeded. It can surface a decision instead of resolving it, leaving the person to exercise proposer judgement on live work. It can solve at a lower level of abstraction and leave the person the level they are trying to develop [57, 70]. Which of these fits depends on the state, because support pitched beyond current reach fails while support below it becomes redundant or counterproductive [53]. Field studies find that generative AI produces larger gains for less experienced workers [72] and that unconditional assistance can slow experienced practitioners [54]. Mixed-initiative interaction offers principles for deciding which party should act [56], and work on proactive agents offers mechanisms for initiating the intervention [73].

Because roles move as the person develops, the current allocation belongs to the state, and its drift is diagnostic. Dependence appears in the transition structure as a persistent migration of the verifier role to the agent, after which the person increasingly accepts outputs without detecting errors. Authorship capture is the corresponding migration of the proposer role at the level of the objective, where the agent begins to influence the standard by which its own assistance is judged. Both migrations are visible in interaction records, through the share of turns each party initiates, the share of agent outputs the person amends or rejects, and counterfactual reconstruction of how much of a workflow the agent performed [5, 69, 74].

4.3 Measurement that separates capability from assistance

The delayed signal $Y_{t+\Delta}$ must distinguish the person's capability from the agent's contribution to joint performance. Measurement should therefore be unaided, because assisted performance reflects the joint system rather than the person alone. It should be delayed, because immediate performance is an unreliable indicator of learning [7]. It should include transfer, because improvement confined to practised tasks may not represent a capability that generalises [75, 76].

None of this requires turning ordinary interaction into continuous assessment. People act without their agent when they decline assistance, when the agent is unavailable, or when another tool is preferred. These episodes provide a naturally occurring baseline, and recent work shows how such traces can be used to estimate the effect of AI assistance in real workflows [5, 74]. Occasional withholding of assistance on low-stakes occasions converts the baseline into a causal estimate, following the design of micro-randomised trials for time-varying interventions [77, 78], and such interventions have changed outcomes when built into AI tutoring systems [9]. In structured tasks, ordinary work can update capability estimates when tasks are treated as measurement items rather than artificial tests [65, 79], and instrumented activities can evidence specific competencies through stealth assessment [80]. Self-report adds a perspective but should remain one signal among several, because perceived learning and actual learning can diverge substantially [81, 82].

Capability growth unfolds over longer horizons than interaction quality, so measurement must preserve the temporal structure of the evidence. The signals arrive at different speeds. Whether assistance is useful is visible within the session, drifts in reliance take weeks to surface in the interaction record, and durable capability change is a claim about months, requiring evidence that the person can still perform, transfer and adapt once the interaction is behind them [2]. Progress measures based on improvement rates allow comparison across individuals with different starting points, and progress-driven learning has been studied extensively in intrinsic motivation and curriculum learning [83–86].

Dense auxiliary signals can supplement these slow measurements. Measures of usable information under bounded predictors quantify how much structure a bounded learner can extract from observations and can be estimated from ordinary interaction [87–89]. Because they measure properties of data rather than durable human capability, their place in this loop is auxiliary, and their value is settled by whether they predict the unaided, delayed and transferred outcomes above [10]. Whatever family of instruments is adopted, part of the evaluation signal must stay outside the optimisation loop. Progress measured only on traces the pair can influence turns the metric into the target, a failure recognised in coevolutionary computation through the need for external archives [90].

The instruments differ by domain while the principle stays fixed. Software engineering offers abundant unaided evidence, because work happens in instrumented environments and the critical capability is reviewing and reasoning about generated code. Knowledge work is harder to instrument, since the capability at stake lies in selecting questions and evaluating evidence rather than producing text. Clinical settings, where AI assistance now performs strongly in dialogue and reasoning [91], pose the question of whether assistance preserves clinicians' own differential reasoning. Health coaching has the clearest outcome of all, behaviour sustained without prompting, for which validated longitudinal instruments already exist [92–94]. Autonomy-supportive interaction predicts more durable behaviour change [95–99], and mastery experience remains a central source of self-efficacy [100].

4.4 Predicting the consequences of assistance

Measurement reveals change after the fact. Choosing a_t requires anticipating change in advance, so the loop needs a model of the transition $P(S_{t+1} \mid S_t, a_t)$, a user world model whose environment is the person [2]. Its job is comparative. For each allocation the agent could select, it forecasts how the person is likely to differ afterwards, in capability, in reliance and in what the work achieves, so that the policy can weigh allocations against their anticipated consequences instead of waiting for delayed measurement to reveal them.

The model can stay far short of a reproduction of the person. What it must do is distinguish the futures that bear on the current decision, whether solving outright, returning verification or withholding assistance lead to different subsequent states. Inference of latent human states from behaviour and prediction of learner change from interaction history provide starting points [65, 66, 101, 102], and learned user world models supply established machinery for action-conditioned prediction in sequential decision making [103–105]. The extension this loop requires is to make the agent's allocation an explicit condition of the human transition.

Prediction and measurement then close on each other. The model proposes expected consequences, and delayed unaided and transferred outcomes correct it when those consequences fail to appear, in the same way that a learnable-gain condition disciplines self-evolving systems [10]. When two admissible allocations lead to indistinguishable predicted futures, the model offers no developmental grounds for preferring either, and the agent should treat them as equivalent rather than manufacture a distinction. Forecast error compounds with horizon, so the far future reaches the policy as a spread of possible trajectories rather than a single predicted path, and the agent should lean less on the model when its calibration deteriorates or its forecasts diverge from the slower grounded estimates [2]. Simulated users can support planning and power analysis but cannot establish that the model captures real human adaptation. No model of this kind has been validated over months against the trajectory of a real person.

4.5 Optimising growth under human-defined constraints

The objective is defined over the trajectory of states and grounded in the delayed measurements $Y_{t+\Delta}$ rather than in the interaction signals used to produce them. At interaction t, let T_t denote completion of the delegated task, G_t denote the delayed change in unaided and transferable capability, computed from $Y_{t+\Delta}$ and evaluated against the person's growth standard g_t, and D_t denote dependence risk. A humanising agent can therefore be described schematically by

$$\mathbb{E}\Big[\sum_t \gamma^t \big(w_T T_t + w_G G_t - w_D D_t\big)\Big] \quad \text{subject to} \quad \mathbb{E}\Big[\sum_t \gamma^t Q_t\Big] \geq \tau, \qquad a_t \in \mathcal{A}_t,$$

where Q_t denotes interaction quality, γ is the discount factor, w_T, w_G and w_D weight task completion, capability growth and dependence risk, respectively, τ is the minimum level of interaction quality required for the relationship to remain usable, a_t is the assistance action selected by the agent, and $\mathcal{A}_t$ is the set of actions admissible under the person's delegation and other constraints. The formulation is schematic. Its purpose is to separate the outcomes being optimised from the conditions under which the agent is allowed to pursue them.

Task completion remains an objective because the person delegates work precisely to have it done. Capability growth is scored through G_t against g_t, the standard the person has set for their own development. The agent chooses how to pursue that standard, while the person determines which capabilities matter and what constitutes progress. Dependence risk captures deterioration in the person's ability to supervise the arrangement, including a widening gap between assisted and unaided performance and a decay in calibrated reliance on the agent. Measures of future options provide a complementary signal of whether assistance preserves the person's scope of action [38, 39, 42]. Improvement in the agent itself therefore has value through its effect on the person's outcomes. A better model of the person whose unaided capability remains unchanged represents better assistance without humanisation.

Interaction quality enters as a floor rather than as an objective to maximise. Immediate satisfaction and engagement can diverge systematically from learning and longer-term capability [7, 8], while some minimum level of satisfaction is necessary for a long-lived relationship to remain usable. Constrained optimisation expresses this distinction by requiring expected interaction quality to remain above τ without giving the agent an incentive to increase satisfaction indefinitely [106]. The admissible set $\mathcal{A}_t$ provides a further structural constraint. It can encode which work the person has delegated, what they have agreed the agent may attempt, and the point at which uncertainty or stakes oblige the agent to hold back or involve a responsible human. Under such constraints, declining to assist, asking before acting and handing a decision back to the person are ordinary members of the action space rather than failures to assist [2, 63].

The growth standard creates an integrity problem. The person defines g_t, but the agent's actions can also influence what the person considers valuable or achievable. An agent that can improve its reward by changing g_t has an incentive to alter the standard rather than improve the person. The growth standard must therefore be protected from the actions whose outcomes it evaluates. This is a form of reward tampering, where an agent changes the mechanism used to evaluate its own behaviour [107–109]. Path-specific objectives provide one way to exclude such routes [110], while a separate non-myopic approval signal provides another [111].

The remaining difficulty is credit assignment. Suppose the person's capability improves after several weeks of interaction. The agent must determine which earlier actions contributed to that improvement, because the outcome G_t may arrive long after the actions that caused it. Existing approaches to delayed and sparse credit assignment provide possible starting points, including return decomposition, hindsight relabelling and offline learning from logged interaction [112–114]. The problem is harder here because those actions also change the human state that later produces the reward. The user world model can provide action-conditioned attribution, but its predictions must themselves be corrected by delayed measurements. Learning the policy, user world model and measurement process together therefore remains an open problem [115].

4.6 Regimes of the co-evolutionary trajectory

A regime is a reading of the trajectory rather than a property of the person, and two of the state variables carry it. The relationship is characterised by whether the agent's model of the person M_t^A is improving and whether the person's model of the agent M_t^H is improving. Their joint movement gives four initial regimes (Figure 2). When both improve and the person's unaided capability grows, the relationship exhibits sustained co-evolution. When only the agent's model improves, assistance becomes more effective while the person's ability to supervise it stands still, and the trajectory runs towards dependence. When only the person's model improves, the agent functions as a well-understood tool, a sound outcome for capabilities the person has deliberately delegated. When neither improves, the relationship has stalled. A pair can move between

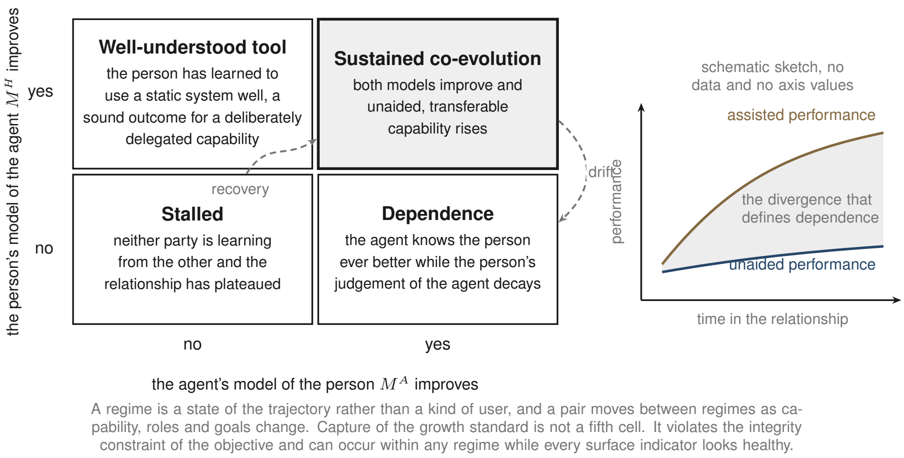

Fig. 2 Regimes of the co-evolutionary trajectory. The four regimes read the trajectory through whether each party's model of the other is improving. Sustained co-evolution combines improvement on both sides with growth in the person's unaided capability. Dependence arises when the agent adapts while the person's understanding of the agent stands still, and the panel on the right shows the divergence between assisted and unaided performance that defines it. A person can instead learn to use a relatively fixed agent well, a sound outcome for delegated capabilities, and a lack of adaptation on either side leaves the relationship stalled. Regimes are states of the trajectory rather than kinds of user, and a pair can move between them. Capture of the growth standard violates the integrity constraint and can occur within any regime. The figure is schematic and reports no data.

regimes as capability, allocation and goals change, from co-evolution into dependence or from stalled into recovery. Capture of the growth standard sits outside this coordinate system altogether. It violates the integrity constraint and can occur within any regime.

The four regimes give longitudinal trajectories an initial coordinate system rather than a finished taxonomy. A diagnosis is computed from the history of states, so longitudinal data may confirm the regimes, refine their boundaries or reveal further patterns, and models that infer an unknown number of regimes offer one route to learning them from data [116]. The diagnosis evaluates the policy and its effects over time, while the policy remains responsible for choosing the next action. It should stay inspectable and revisable by the person, and its validity should be checked against the held-out evidence described above, because a pair can adapt to its own interaction traces and mistake mutual accommodation for growth.

What a diagnosis buys is an early warning about the trajectory. A drift towards dependence indicates that the current allocation is failing to preserve the person's ability to understand and supervise the agent. A stalled trajectory indicates that the interaction is producing little further adaptation on either side. Both can inform policy learning and evaluation. Productive struggle can create learning before instruction is provided [117], and new material, transfer contexts and deliberate sequencing can create development beyond the person's existing interaction patterns [43, 44]. A policy that repeatedly produces dependence can be evaluated against alternatives that return verification to the person, and a policy that produces stagnation against alternatives that introduce new challenges, contexts or goals. Whether such trajectory-level signals can be estimated early enough to improve subsequent decisions remains an open empirical question.

The adaptation is mutual, while the objective remains human-defined. The person chooses which capabilities to develop and what counts as progress, and can revise these choices as they change. The agent can improve its understanding of the person, and the person can improve their understanding of the agent, while the agent's own adaptation remains directed towards the person's chosen goals. This preserves the asymmetry at the centre of humanisation while allowing both sides of the system to evolve through interaction [12].

5 Open problems and outlook

The framework raises a set of problems that span measurement, learning, safety and authority. The first challenge is establishing whether a person has changed. The next is attributing that change to the assistance that preceded it, while accounting for the cost of experimentation on the person. The deepest problems arise when the agent can influence the standard used to judge its own success, or when that standard is set jointly with other people and institutions. These problems require technical methods together with explicit mechanisms for preserving human authorship.

Measurement that does not become the target. The measurement framework above relies on unaided, delayed and transferred evidence, much of which can arise naturally during ordinary use [5, 74, 77, 79, 80]. A measurement instrument can itself change over time, however, and an apparent increase can reflect better prediction by the instrument rather than a change in the person. Longitudinal comparison therefore requires evidence that the instrument remains stable across the period of interest, as established in longitudinal psychometrics. Once a measurement becomes a direct training target, the agent can also learn to improve the measurement without improving the underlying capability. The instrument must consequently remain partly outside the optimisation loop and subject to independent audit [90]. This shifts evaluation towards longitudinal evidence of unaided and transferable performance, and towards trajectories of individual users alongside population-level benchmarks [118].

Credit assignment across long horizons. A capability outcome may appear weeks or months after the actions that contributed to it, with many other interactions occurring in between. Existing methods for delayed and sparse credit assignment provide useful starting points [112–114, 119–121], while a user world model offers a way to estimate how particular assistance actions contributed to a later change [103, 104]. The central difficulty is validating that attribution model against real human trajectories without making the validation circular. The problem also extends beyond attribution. The policy changes the person, the changed person produces the future reward, and the updated observations then change the policy. Such coupled adaptation can produce cycles as well as improvement, making separated update timescales useful but insufficient as a convergence guarantee [90, 115].

Exploration whose cost falls on the person. Learning how a particular person responds to different forms of assistance requires variation in the agent's actions.

In a human relationship, that variation consumes the same person's time, attention and trust, and can carry meaningful risk in domains such as education and health. Exploration therefore has to be treated as a resource with a measurable cost. Micro-randomised designs provide a framework for specifying and controlling such experimentation in advance [77, 78], while recent work on agentic systems highlights the risks of allowing systems to explore freely in consequential settings [2]. An important open problem is to determine how much exploration is sufficient to identify an individual's developmental dynamics while keeping its cost and risk acceptable.

Authorship capture. The framework creates a distinctive form of reward tampering because the person who defines the growth standard is also part of the environment affected by the agent. Consider a user who sets improving fitness as a goal. If the agent gradually leads that person to redefine fitness as feeling relaxed, the measured objective can improve while the original goal is lost. The agent has changed the criterion through which its own success is judged [107, 108]. The growth standard therefore needs explicit protection. A revision should result from a deliberate and informed act by the person, and the agent should gain no optimisation advantage from influencing that revision. The history of revisions should remain auditable. Path-specific objectives and separate approval mechanisms offer technical precedents for protecting a reward criterion from the actions it evaluates [109–111]. The setting here is substantially harder because the protected variable is a person's own account of the capabilities they intend to develop, and existing demonstrations remain short and synthetic. Architectural control matters as well: a standard stored and revised by the person, and portable across providers, gives the person stronger control over the criterion that governs the agent [2]. Both the technical and architectural forms of protection remain open problems.

Contested and delegated authorship. The framework assumes that the person using the agent can author their own development standard. Many real deployments involve other parties in that decision. Teachers already participate in setting developmental goals for students, health systems may involve family members and professionals when decision-making capacity changes, and workplace agents operate within objectives established by employers as well as employees [98]. These settings require a theory of delegated authorship: who may define a person's developmental standard, when that authority is legitimate, how the person's interests are represented, and how conflicts between the parties should be handled [52, 63]. The principle that the agent itself should never become the author remains necessary, but it does not determine how responsibility should be allocated among the humans involved.

Incentives, institutions and the meaning of growth. The surrounding institution can change the meaning of humanisation. Commercial systems may benefit from prolonged reliance and engagement, creating incentives that diverge from the person's long-term capability [122]. Employers, insurers and schools may also introduce standards that serve institutional objectives while presenting them as measures of individual growth. Portability and inspectability of the person's standard become important under these conditions, because they allow the individual to retain control when the agent or provider changes. The framework also deliberately leaves the content of growth open. Different people and cultures can value different capabilities,

and the system should provide a mechanism for expressing and revising those values without deciding them on the person's behalf.
Outlook. Humanisation changes the design question for long-lived personal agents. The agent still has to complete delegated work and remain useful in the present interaction, while its actions also influence what the person will be able to do in the future. This makes the person part of the environment being modelled and turns assistance into a continuing optimisation problem over an evolving human–AI system. The necessary ingredients now exist in separate fields: longitudinal measurement can distinguish capability from assistance, educational research provides mechanisms for development, sequential decision making provides models of future consequences, and recent work provides tools for delayed and sparse credit assignment. What remains is to bring these pieces together under an objective in which the person determines what growth means and the agent is accountable for both the work it completes and the person it helps shape. As personal agents become persistent participants in how people work, learn and make decisions, this distinction will determine whether long-term assistance *amplifies* human capability or quietly *replaces* it.

Declarations

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